

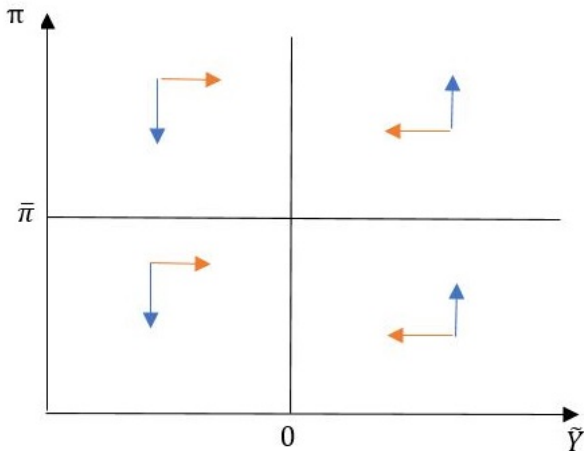
ECON 311

Lecture 18: Financial Crisis and Covid Recession

Using the AS-AD Model

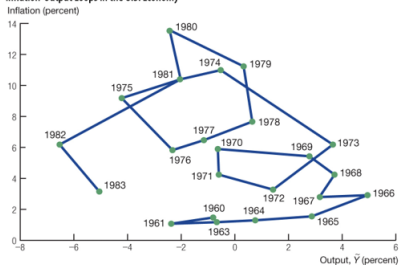
Financial Crisis of 2008 and Covid Recession

Phase diagram cost push and demand shocks

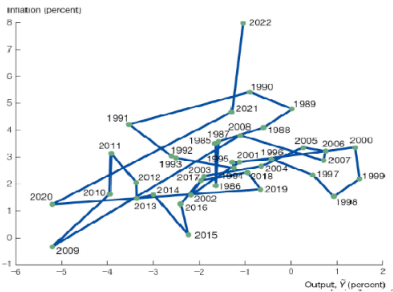


Empirical Evidence

Inflation-Output Loops in the U.S. Economy



(a)



(b)

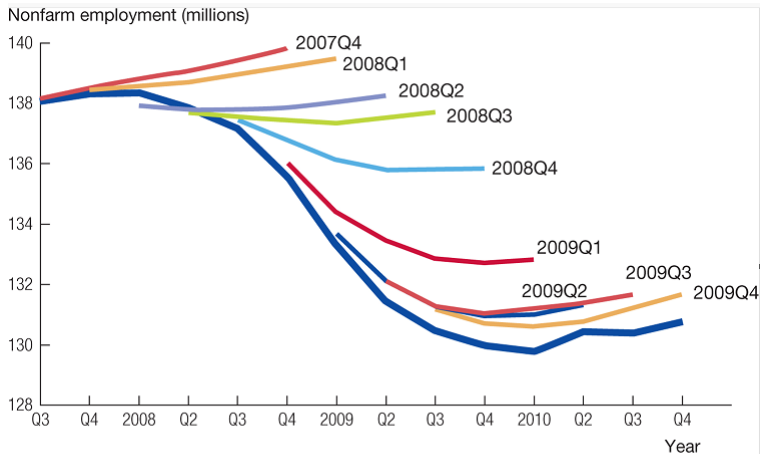
Rules vs. Discretion

- Explicit Policy Rules:
 - ▶ (pros) Help anchor inflation expectations and maintain a stable inflation environment where shocks are deflected quickly
 - ▶ (cons) To work, policy rules must be credible, and, therefore, provide little flexibility to deviate from targets
- Discretionary Monetary Policy:
 - ▶ (pros) Provides flexibility to pursue expansionary economic policy
 - ▶ (cons) Shocks can be very disruptive as inflation pressures do not dissipate quickly
- It is extremely important that, in Modern Economies, CBs are independent from the government in power

Using the AS-AD Model

Financial Crisis of 2008 and Covid Recession

Great Recession



Source: James H. Stock and Mark W. Watson, "Twenty Years of Time Series Econometrics in Ten Pictures," *Journal of Economic Perspectives* (Spring 2017).

Copyright © 2021 W.W. Norton & Company

Great Recession: Shocks

Sequence of events that led to the crisis

- Global Savings Glut
 - ▶ In 1996, developing countries borrowed from the rest of the world.
 - ▶ In 2003, developing countries were saving a net \$205 billion into the world's capital markets.
 - ▶ Most of these savings enter the US market
- Housing Market and Subprime lending
- Rise interest rates and decrease housing prices
- Oil Prices

Great Recession: Shocks

Sequence of events that led to the crisis

- Global Savings Glut
- Housing Market and Subprime lending
 - ▶ Free Money led to subprime lending in housing markets through a variety of financial instruments deemed as "safe" assets (securitization)
 - ▶ The system worked as long as housing prices were increasing
- Rise interest rates and decrease housing prices
- Oil Prices

Great Recession: Shocks

Sequence of events that led to the crisis

- Global Savings Glut
- Housing Market and Subprime lending
- Rise interest rates and decrease housing prices
 - ▶ Between 2004 and 2006, the Fed raised its interest rate from 1.25% to 5.25% $\Rightarrow$ Higher interest rates ($>$ teaser) for subprime debtors
 - ▶ From mid-2006 to the beginning of 2012, Housing prices plummeted by 36% (bubble burst)
- Oil Prices

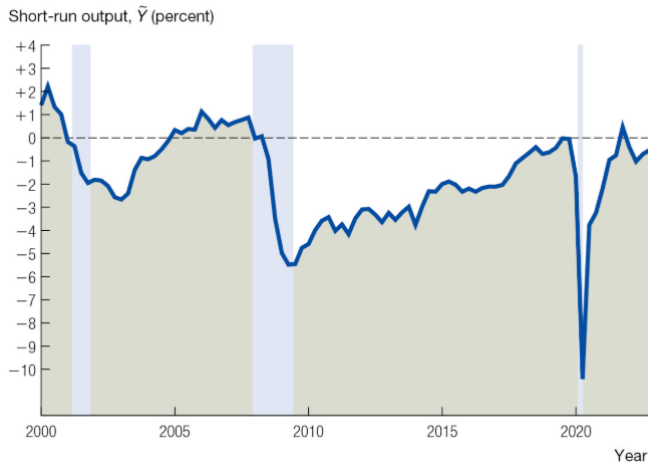
Great Recession: Shocks

Sequence of events that led to the crisis

- Global Savings Glut
- Housing Market and Subprime lending
- Rise interest rates and decrease housing prices
- Oil Prices
 - ▶ In 2002, oil was about \$20 per barrel.
 - ▶ During the summer of 2008, prices peaked at \$140 per barrel.

Great Recession

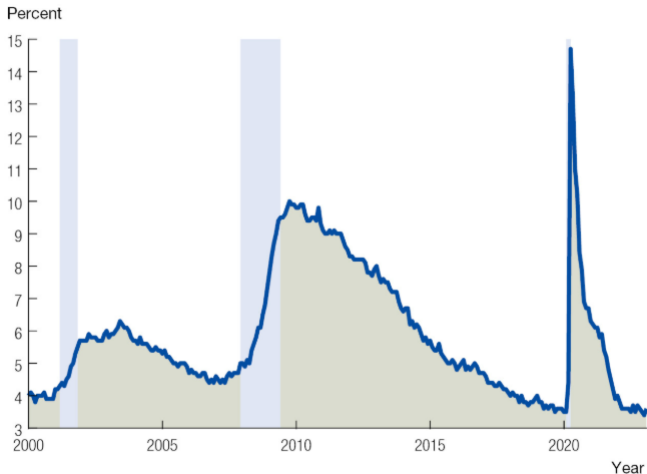
U.S. Short-Run Output, $\tilde{Y}$



Source: The FRED database and author's calculations.

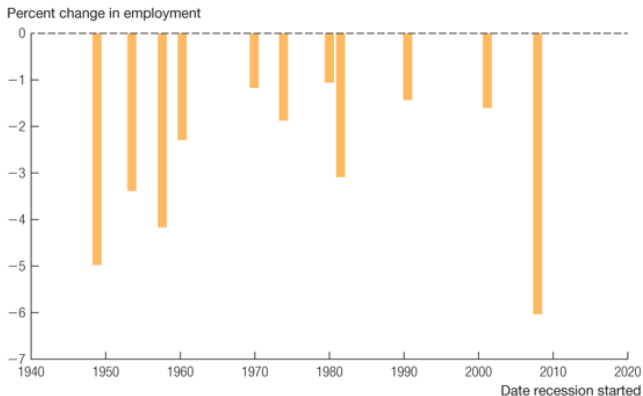
Great Recession

The U.S. Unemployment Rate



Source: The FRED database.

Great Recession

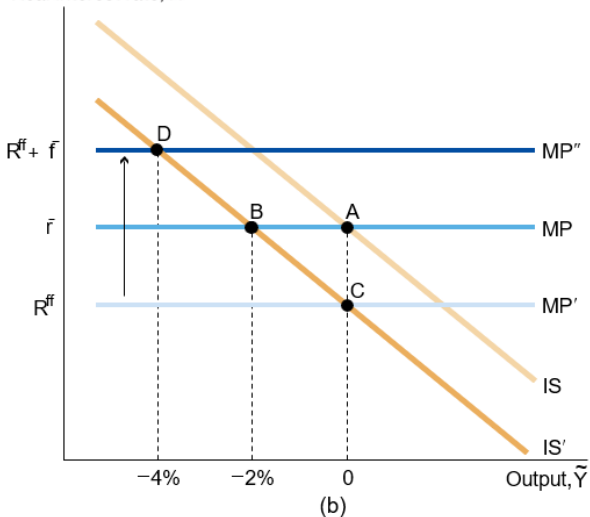


Source: U.S. Bureau of Labor Statistics, Civilian Unemployment Rate [UNRATE], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/UNRATE>, September 22, 2019.
Copyright © 2021 W.W. Norton & Company

Great Recession

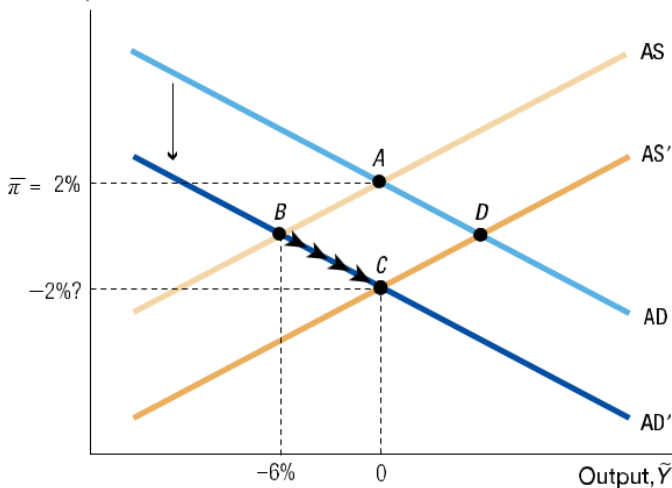
A Housing Bubble and a Financial Crisis

Real interest rate, R



Great Recession

The Financial Crisis in the AS/AD Framework
Inflation, π



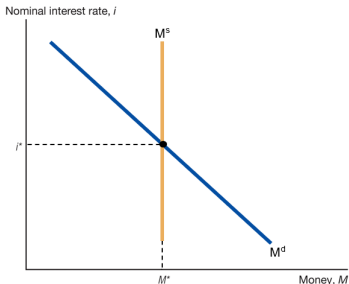
Monetary Policy— Revisited

- $R_t = i_t - \pi_t$, with sticky inflation:

changes in $i_t \Rightarrow$ changes in R_t

- How does the Central Bank set i_t :

1. Overnight rate
2. Open Market Operations:



Liquidity Trap

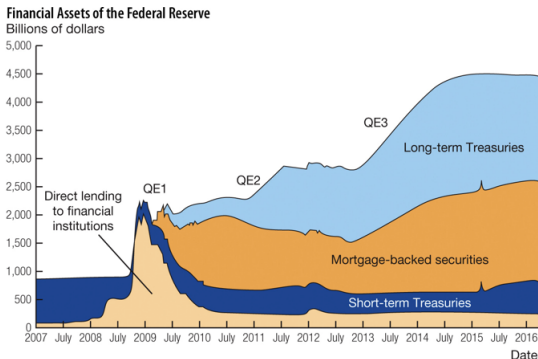
- i_t cannot be less than zero, so the lowest R_t can be is:

$$R_t = -\pi$$

- When there is deflation (negative inflation) as in 2008/9, this severely restricts the ability of the government to perform MP.
- Government needs to resort to non-traditional monetary policy instruments, e.g:
 1. Quantitative easing
 2. Forward Guidance

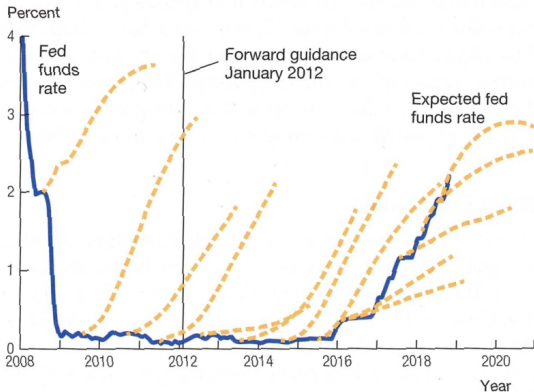
Quantitative Easing

- **Quantitative Easing:** Program in which the CB buys other assets than government short-term bonds
 - ▶ During the Financial Crisis these assets were troubled assets directly injecting liquidity to financial institutions and allowing them to continue lending



Forward Guidance

- **Forward Guidance:** CB signalling commitment to certain monetary policy in the near future
 - ▶ Goal of this policy is to influence expectations and, therefore, behaviour of investors

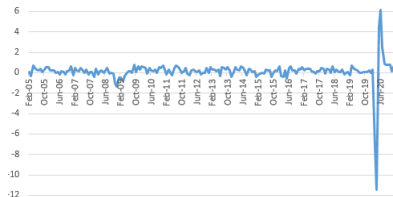


Economic Indicators during Covid: RGDP

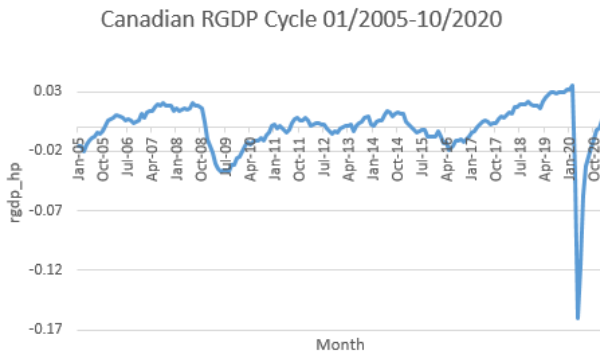
Canadian RGDP (Chained 2012 Dollars)
1/2005-10/2021



% Change in RGDP

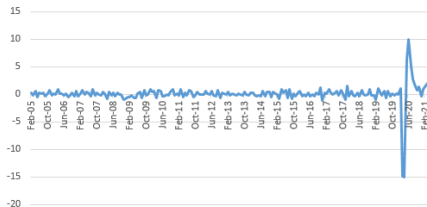


RGDP– Cycle

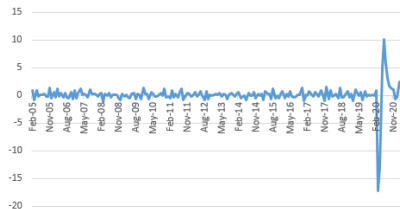


Employment in Canada (Total Hours Worked)

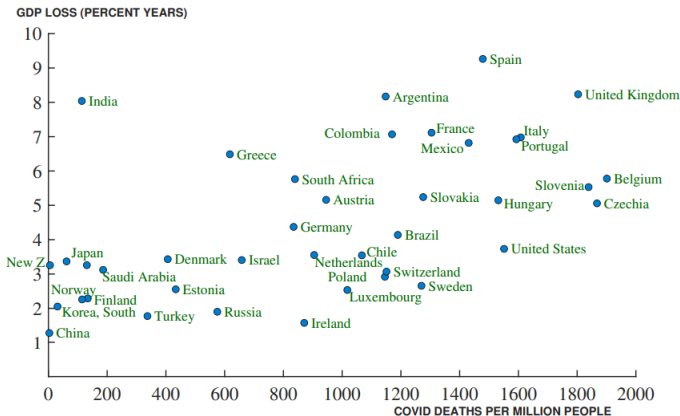
% Change in Total Hours Worked (all sectors)



% Change in Total Hours Worked (service)



International Covid Shock



Note: "GDP Loss" is the *cumulative* loss in GDP since the start of 2020 and is annualized. For example, a value of 6 means that the loss since the start of 2020 is as if the economy lost six percent of its annual GDP. GDP data are through 2020Q3 and COVID-19 deaths are as of February 26, 2021.

COVID Recession

- Through the lens of our model, the COVID Recession is hard to analyse because:
 - ▶ Multiple Shocks to the economy (supply-side and demand-side)
 - ▶ (Short-run???) shocks to production capacity
 - ▶ Our framework allows for either short-run supply shocks to the firm's price setting behaviour (cost-push shocks) or long-run shocks to the production capacity.
 - ▶ Consensus is that it affected $\bar{Y}$ in the short-run, which affected demand parameters in the long-run with the short-run adjusting right away (did not see any deflation pressure — unlike overshooting in the IS-MP model)
 - ▶ As we recover, agents are failing to see that this is a more permanent effect on $\bar{Y}$, which is why we are currently seeing short-run overshooting.
 - ▶ In addition, we are experiencing a cost-push shock due to the war in Ukraine.